Physics-101, Mid-semester examination

Department of Physics

Shiv Nadar University (SNU), Gautam Budhha Nagar

Time: 90 Minutes (9:30-11:00 AM./30th Sep 2016) Max. Marks: 25.

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1. A particle of mass m is set to move along an inclined plane of slope angle α from horizontal at the point O with a velocity u as shown in figure below (Fig1). The coefficient of friction (μ) between the particle and the plane is speed dependent, $\mu = k\,\dot{r}$ where r is the distance of the particle from origin. The length of the ramp is s. After that the particle flies in air as a projectile.

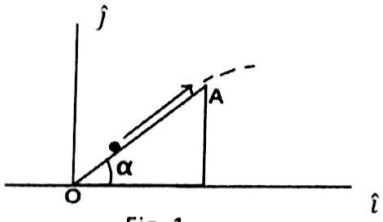
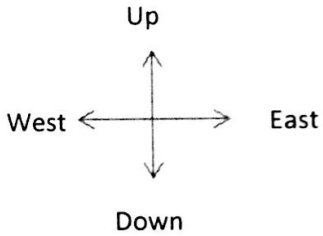


Fig. 1



Give the answer of the following questions.

(i) Draw Free body diagrams of the particle during the journey from O to A and after leaving the point A. [1]

(ii) What is the velocity of particle as a function of time from O to A? [3]

(iii) What is the position of particle as a function of time from O to A? [2]

2. A block of mass m rests on a wedge (mass M) inclined at angle θ. The coefficient of friction between the block and wedge is μ. Do the analysis from inertial frame as shown in the figure (Fig.2) . **Do not use any other frame of reference.**

(a) The wedge is given horizontal acceleration a, by applying a force F. Assuming that $\tan\theta > \mu$, draw neat free body diagrams of the block and wedge for minimum value of F and maximum value of F so that the block do not slip on the wedge.[4]

(b) Find the **maximum** Force F so that the block do not slip on the wedge. [2]

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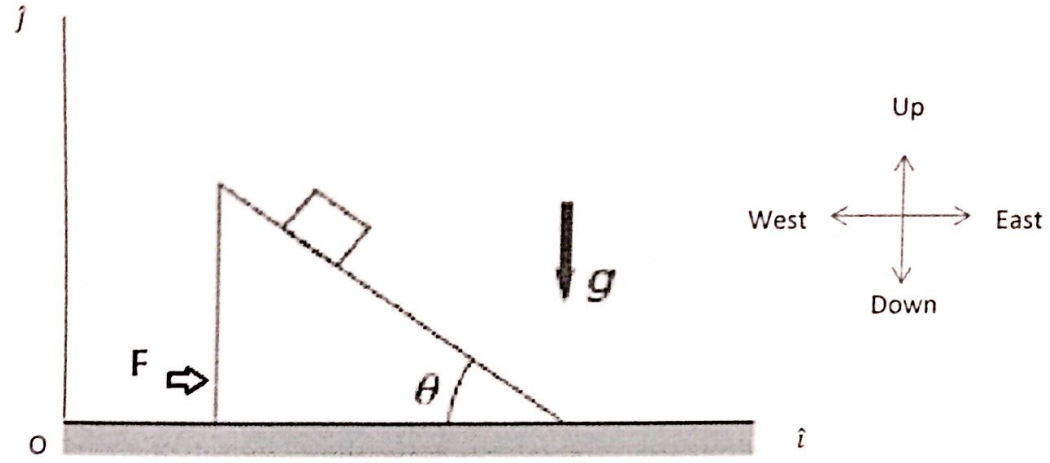

Fig. 2

3. For a simple pendulum while deflected at angle θ from vertical, find out equation of motion using polar coordinate system as shown in figure (Fig-3). Show that it does a simple harmonic motion for small θ. [5]

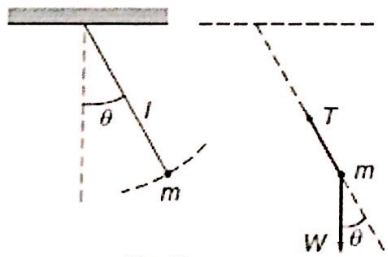

Fig. 3

4. A particle of mass m moves in one dimension along the positive x axis. A constant force of attraction directed toward the origin with magnitude B and a repulsive force with magnitude A/x^2 acts on the particle.
   (a) Find the equilibrium point of the particle. [2]
   (b) What is the frequency of small oscillations about the equilibrium point? [3]
   Hint : $If\ y \ll 1, (1+y)^n \approx 1 + ny$
   (c) If in addition to restoring force a velocity dependent force $(-bv)$, where v is the instantaneous velocity, also act on the particle around equilibrium point. Write down the differential equation of the particle for small displacement.[2]
   (d) How much time will it take to 90% drop in the amplitude of vibration. [1]

$\ddot{x} = \omega_0^2 x$

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